Sandbox Mini-App Approach

CodeFixPanda_Kotlin · senior Android interview preparation · 2026-11-11 · build 64.4

Purpose

The real app is large. To absorb it, each concept from the code walkthrough gets a tiny, runnable example in a separate sandbox Activity, `MainActivity2`. The real code stays untouched, and the sandbox is never merged into `develop`.

Branches

These follow the branch conventions document. Both branches were cut from `develop` at 95ed21d, build 63.9, last merged PR #61.

Branch	Role	Merge?
`feature-set/codeWalkthrough`	Staging. Stays at the `develop` tip. It moves up to the step's head only when Gene says so.	No PR is opened
`sandbox/miniAppDemos`	Step branch. Every commit is made here, and we stay on it until Gene says otherwise. It is named `sandbox/` instead of `refinement/` because it is not a refinement.	Never merged

The earlier branches `sandbox/walkthrough-mini-app`, `sandbox/walkthrough-step-02-koin-lazy-single` and the session branch `claude/festive-mayer-purx3h` and the first step branch `refinement/codeWalkthrough-koin` are abandoned. Gene deletes them.

What the step branch changes

- **New file** `app/src/main/java/io/github/genesjc/codefixpanda/MainActivity2.kt`, the mini app.
- **Edited** `app/src/main/AndroidManifest.xml`. The launcher `<intent-filter>` moved from `MainActivity` to `MainActivity2`. `MainActivity` is otherwise unchanged.
- **Edited** `app/build.gradle.kts`. The build number is 64.n, one step per commit.
- **Docs** in `_meta/docs`: the conventions PDF and this report.
- `CodeFixPandaApplication` still runs for both Activities, so Koin is started and the sandbox can use the real dependency injection setup.

Switching between the sandbox and the real app

Move this block between the two `<activity>` elements in the manifest, then rebuild:

```
<intent-filter>
    <action android:name="android.intent.action.MAIN" />
    <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
```

Under `MainActivity` the real app runs. Under `MainActivity2` the sandbox runs.

Working method for each chunk

1. Walk one chunk of real code, following the conventions PDF.
2. Where it helps, add a minimal example of the same concept to `MainActivity2.kt` as one commit on the step branch, with the build number raised and shown at the end of the commit subject.
3. Run it and compare what happens with what the real code does.

Progress

Chunk	Real code	Concept	Sandbox demo
1	`MainActivity.onCreate`	Activity, `enableEdgeToEdge`, `setContent`, recreation	Committed in 64.4, not yet run
2	`CodeFixPandaApplication.onCreate`	Koin starts once per process, and a `single` is built lazily	An earlier version is in commit aeaba46 (64.1). It returns when we reach Chunk 2.

Next: Chunk 2, `CodeFixPandaApplication.onCreate`, then `CodeFixPandaApp()` in `ui/CodeFixPandaApp.kt`, the root composable that `setContent` calls.

The Chunk 1 demo

`MainActivity2` repeats the five lines of `MainActivity.onCreate`, with a log line and a plain `Text` as the content. It leaves out the manifest's `android:configChanges` on purpose. Expected behavior, from reading the code and not yet observed:

- Logcat tag `Mini` prints `onCreate: fresh launch = true` on a fresh start.
- Rotate the device. The Activity is recreated, `onCreate` runs again and the log prints `false`. The real `MainActivity` avoids this with `configChanges`.
- The `Text` is at the top of the screen and draws under the status bar because of `enableEdgeToEdge()`. Comment that line out and compare.

Known limits

- **Not compiled yet.** The session container has no Android SDK, so the demo has not been built or run. The manifest is valid XML. The expected Logcat order is what the code should produce, not an observed result.
- CI runs only for pushes and PRs on `develop` and `main`, so these branches get no CI run. The blank-line guard rules were applied by hand.
- Because the manifest is edited to switch Activities, move the launcher filter back before running the real app.

For another conversation joining this work

1. Read `_meta/docs/code-walkthrough-conventions.pdf` and the branch conventions document.
2. Check out `sandbox/miniAppDemos`.
3. Follow the conventions. Do not merge the step branch or open a PR from the staging branch.